

Skill 9:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>

/* =====
Built-in command functions
===== */

void builtin_pwd()
{
    char cwd[PATH_MAX];

    if (getcwd(cwd, sizeof(cwd)) != NULL)
    {
        printf("Current directory: %s\n", cwd);
    }
    else
    {
        perror("pwd");
    }
}

/* Change directory */

void builtin_cd(char *path)
{
    char previous[PATH_MAX];

    /* Store previous directory */

    if (getcwd(previous, sizeof(previous)) == NULL)
    {
        perror("getcwd");
        return;
    }
}
```

```

    /* Change directory */

    if (chdir(path) != 0)
    {
        perror("cd");
        return;
    }

    printf("Previous directory: %s\n", previous);
    printf("Directory changed successfully.\n");

    builtin_pwd();
}

/* Echo command */

void builtin_echo(char *text)
{
    if (text != NULL)
        printf("%s\n", text);
}

/* =====
   Dispatch table
   ===== */

typedef void (*builtin_function)(char *);

typedef struct
{
    char *name;
    builtin_function function;
} Builtin;

```

```

Builtin builtins[] =
{
    {"cd", builtin_cd},
    {"pwd", (builtin_function)builtin_pwd},
    {"echo", builtin_echo}
};

int builtin_count = 3;

/* =====
   Execute built-in command
   ===== */

void execute_builtin(char *command, char *argument)
{
    for (int i = 0; i < builtin_count; i++)
    {
        if (strcmp(command, builtins[i].name) == 0)
        {
            printf("Executing built-in: %s\n", command);

            if (strcmp(command, "pwd") == 0)
            {
                builtin_pwd();
            }
            else
            {
                builtins[i].function(argument);
            }

            return;
        }
    }

    printf("Invalid command: %s\n", command);
}

```

```
/* Invalid path */

printf("\n4. Testing invalid path:\n");
execute_builtin("cd", "/this/path/does/not/exist");

/* Return to previous known directory */

printf("\n5. Returning to home directory:\n");
execute_builtin("cd", getenv("HOME"));

/* Echo test */

printf("\n6. Testing echo built-in:\n");
execute_builtin("echo", "Hello from built-in command");

/* Invalid command */

printf("\n7. Testing invalid command:\n");
execute_builtin("unknown", NULL);

/* Final state */

printf("\n8. Final working directory:\n");
execute_builtin("pwd", NULL);

printf("\n=====\n");
printf("PRACTICAL COMPLETED\n");
printf("=====\n");

return 0;
}
```

```
bhanu@DESKTOP-DE32HQ8:~$ cd ~
mkdir -p ~/cd_builtin
cd ~/cd_builtin
bhanu@DESKTOP-DE32HQ8:~/cd_builtin$ pwd
/home/bhanu/cd_builtin
bhanu@DESKTOP-DE32HQ8:~/cd_builtin$ nano cd_builtin.c
bhanu@DESKTOP-DE32HQ8:~/cd_builtin$ gcc cd_builtin.c -o cd_builtin
bhanu@DESKTOP-DE32HQ8:~/cd_builtin$ ls
cd_builtin  cd_builtin.c
bhanu@DESKTOP-DE32HQ8:~/cd_builtin$ ./cd_builtin
=====
DIRECTORY NAVIGATION AND BUILT-INS
=====

1. Initial working directory:
Executing built-in: pwd
Current directory: /home/bhanu/cd_builtin

2. Changing directory to /tmp:
Executing built-in: cd
Previous directory: /home/bhanu/cd_builtin
Directory changed successfully.
Current directory: /tmp

3. Changing directory to /:
Executing built-in: cd
Previous directory: /tmp
Directory changed successfully.
Current directory: /

4. Testing invalid path:
Executing built-in: cd
cd: No such file or directory

5. Returning to home directory:
Executing built-in: cd
Previous directory: /
Directory changed successfully.
Current directory: /home/bhanu

6. Testing echo built-in:
```

```
6. Testing echo built-in:  
Executing built-in: echo  
Hello from built-in command
```

```
7. Testing invalid command:  
Invalid command: unknown
```

```
8. Final working directory:  
Executing built-in: pwd  
Current directory: /home/bhanu
```

```
=====  
PRACTICAL COMPLETED
```

```
=====  
bhanu@DESKTOP-DE32HQ8:~/cd_builtin$ |
```